

Lecture - 3 - Approx/ Estimation Error & ERM.

CS229 - Learning Theory

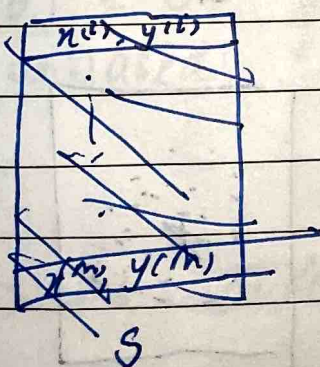
- Setup / Assumptions
- Bias & Variance
- Approx Estimation
- Empirical Risk Minimizer
- Uniform convergence
- VC dimension

Assumptions

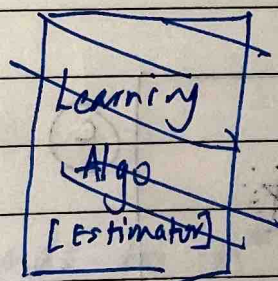
- 1) Data distribution D
 $(x, y) \sim D$ - train
 $(x, y) \sim D$ - test

We have a data generating process from where we have a training & test set which are from same distribution

- 2) All samples are independent.



$\sim D \Rightarrow$

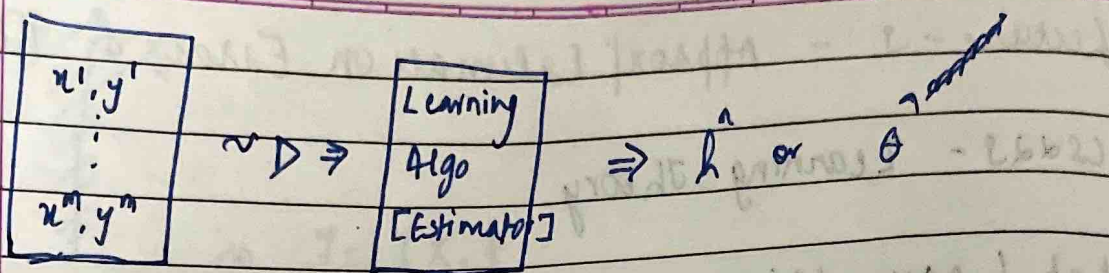


$\Rightarrow$

$\hat{h}$ or $\hat{a}$
sampling distribution

Random

Deterministic function



Random
Variable

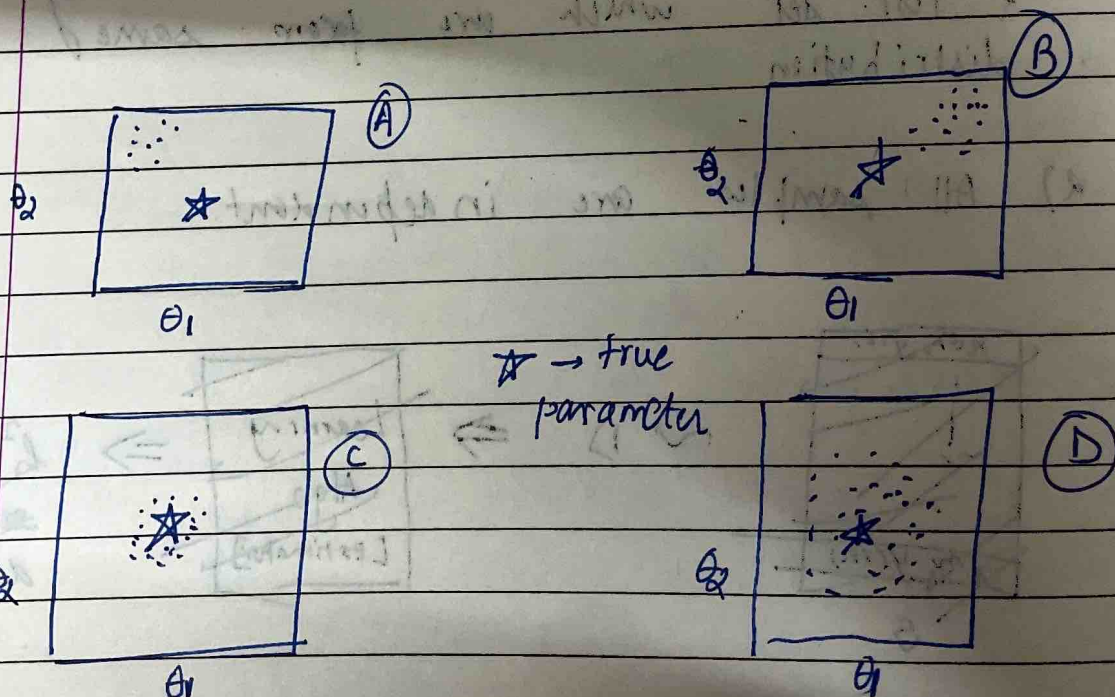
Deterministic

Random Variable

The distribution of $\hat{\theta}$ is called sampling distribution.

θ^* or h^* is a "true" parameter, is a const we don't know, & it is not random.

Parameter View



Bias - is the sampling spread across the true parameter.

Variance - How dispersed the sampling is

B & D - high variance
 C & D - low bias
 A & B - high bias
 A & C - low variance

Bias & Variance are just properties of first & second moments of the distribution.

first moment - big mean
 second moment - variance

If we increase the number of ^{examples} parameters, the variance of $\hat{\theta}$ would become small.
 so as $m \rightarrow \infty$, variance $\rightarrow 0$.

$m \rightarrow \infty$
 $\text{Var}[\hat{\theta}] \rightarrow 0$ "statistical efficiency" rate at which $\text{var}[\hat{\theta}] \rightarrow 0$ as $m \rightarrow \infty$

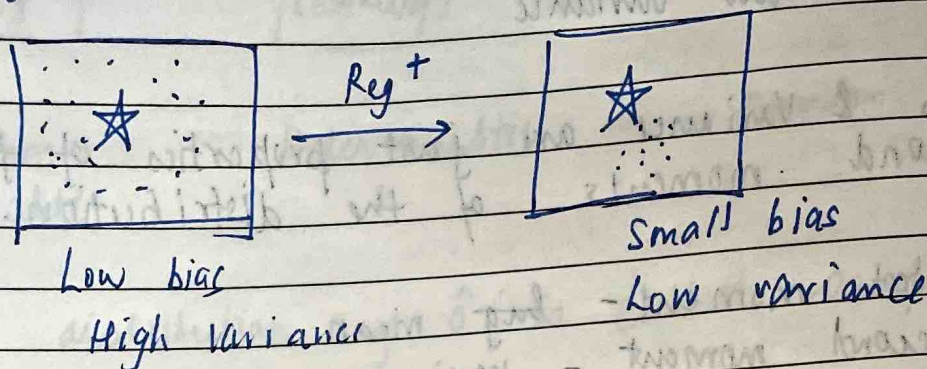
$\hat{\theta} \rightarrow \theta^*$ as $m \rightarrow \infty$ consistent

$E[\hat{\theta}] = \theta^*$ for all m , then it's called unbiased estimator.

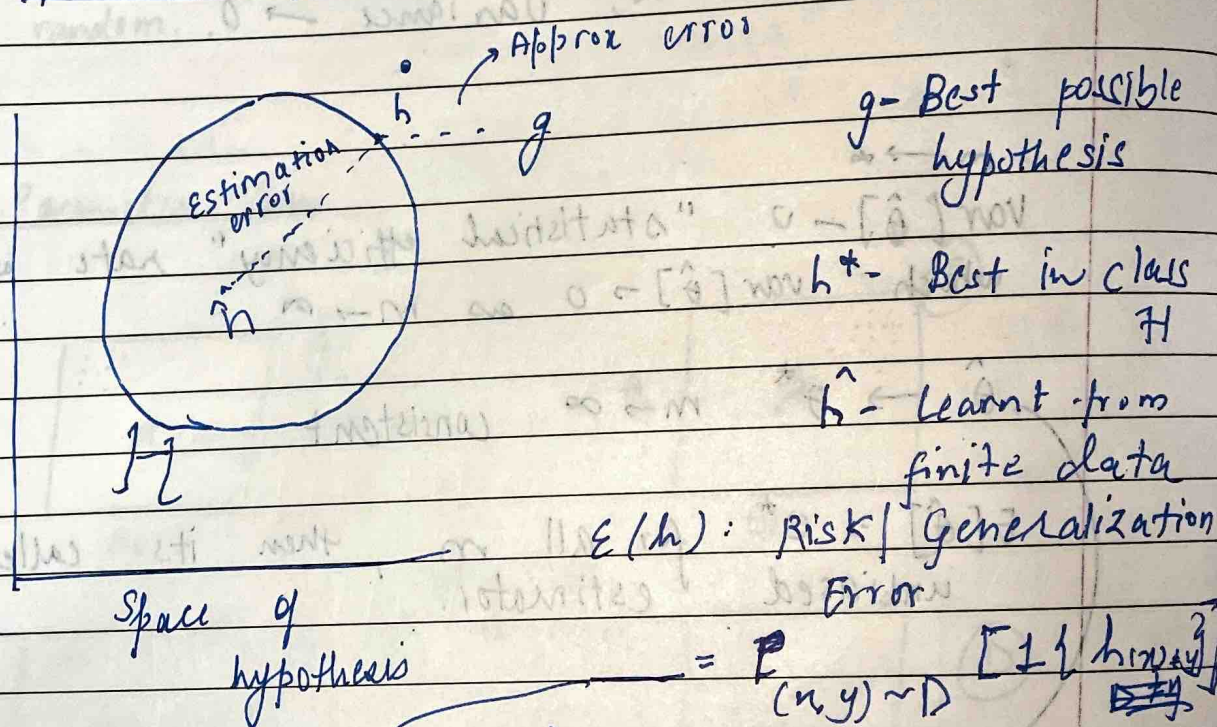
→ The distribution tends to θ^* (const) as $m \rightarrow \infty$

Fighting Variance

- (i) $m \rightarrow \infty$, just get more data.
(ii) Regularization



space of hypothesis



$\hat{E}_S(h)$ - Empirical Risk

$$= \frac{1}{m} \sum_{i=1}^m 1 - [h(x^{(i)}) \neq y^{(i)}]$$

This means the generalization error is the probability that, if we draw a new example (x, y) from distribution D , it will misclassify it.

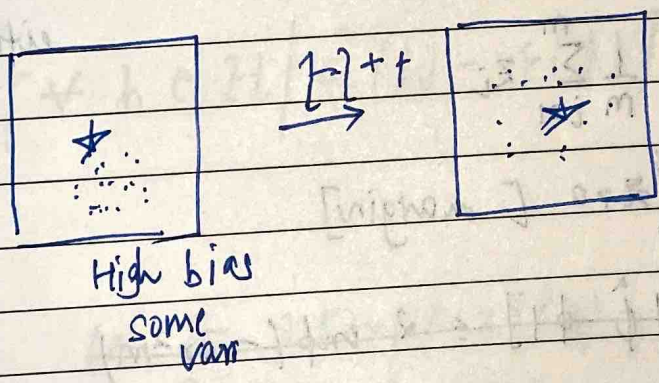
$\mathcal{E}(g)$ = Bayes Error | Irreducible error
 $\mathcal{E}(h^*) - \mathcal{E}(g)$ = Approximation error of class
 $\mathcal{E}(h) - \mathcal{E}(h^*)$ = Estimation error of data

$\mathcal{E}(h) = \text{Estimation error} + \text{Approx error} + \text{Irreducible error}$

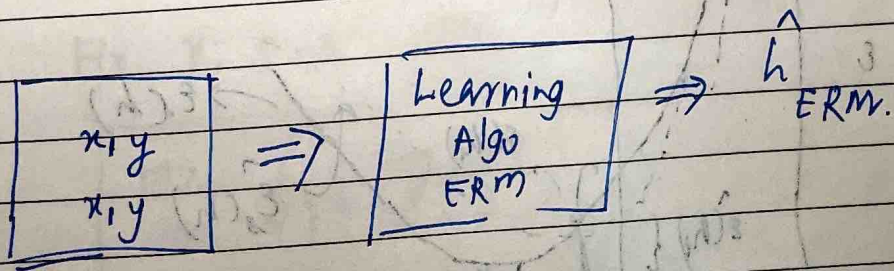
$\mathcal{E}(h) = \underbrace{\text{Est Var}}_{\text{Var}} + \underbrace{\text{Est Bias} + \text{Approx error} + \text{Irreducible error}}_{\text{Bias}}$

Fight High Bias

(i) make $\mathcal{H}$ bigger



Empirical Risk Minimizer (ERM)



$$\hat{h}_{ERM} = \arg \min_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \mathbb{1} \{ h(x_i) \neq y_i \}$$

Uniform Convergence

- (i) $\hat{E}(h)$ vs $E(h)$
- (ii) $\hat{E}(\hat{h})$ vs $E(h^*)$

Tools

- (i) Union Bound

$$P(A_1 \cup A_2 \cup A_3 \dots A_n) \leq P(A_1) + P(A_2) + \dots + P(A_n)$$

- (ii) Hoeffding's Inequality

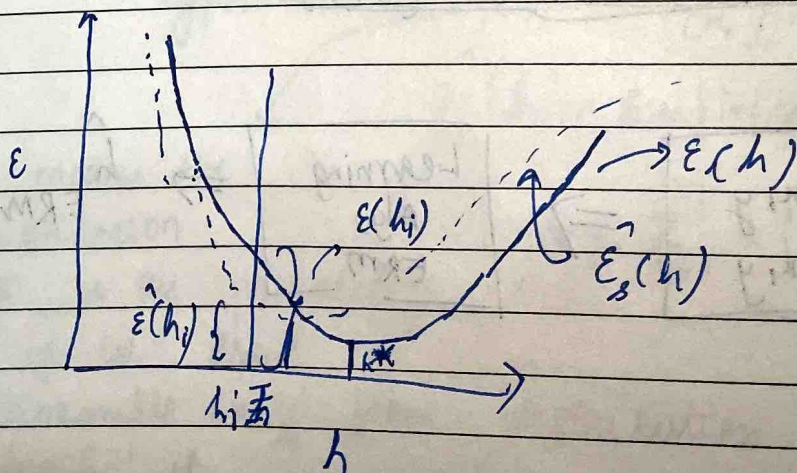
Let $z_1, z_2, \dots, z_m \sim \text{Bern}(\phi)$ each z_i are either 0 or 1.

$$\hat{\phi} = \frac{1}{m} \sum_{i=1}^m z_i$$

Let $\gamma > 0$ [margin]

$$P(|\hat{\phi} - \phi| \geq \gamma) \leq 2 \exp(-2\gamma^2 m)$$

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$$E[\hat{\epsilon}(h_i)] = \epsilon(h_i)$$

Applying Hoeffding's Inequality

$$P[|\hat{\epsilon}(h_i) - \epsilon(h_i)| > \gamma] \leq 2 \exp(-2\gamma^2 m)$$

Finite hypothesis classes

$$|\mathcal{H}| = K$$

$$P[\exists h \in \mathcal{H} \mid \hat{\epsilon}_S(h) - \epsilon(h) > \gamma] \leq K \cdot 2 \exp(-2\gamma^2 m)$$

$$P[\forall h \in \mathcal{H} \mid \hat{\epsilon}_S(h) - \epsilon(h) \leq \gamma] \geq \underbrace{1 - 2K \exp(-2\gamma^2 m)}_{\delta}$$

$$\text{Let } \delta = 2K \exp(-2\gamma^2 m)$$

δ - prob. of error

γ - margin of error

m - sample size

Fix $\gamma, \delta > 0$

$$m \geq \frac{1}{2\gamma^2} \log\left(\frac{2K}{\delta}\right)$$

sample complexity.

$$\varepsilon(\hat{h}) \leq \hat{\varepsilon}(\hat{h}) + \gamma$$

$$\varepsilon(\hat{h}) \leq \hat{\varepsilon}(\hat{h}) + \gamma$$

$$\varepsilon(\hat{h}) \leq \varepsilon(h^*) + 2\gamma$$

with probability of $1 - \delta$, training size m

$$\varepsilon(\hat{h}) \leq \varepsilon(h^*) + 2\sqrt{\frac{1}{2m} + \log \frac{2K}{\delta}}$$

For VC dimension

$$\varepsilon(\hat{h}) \leq \varepsilon(h^*) + O\left(\sqrt{\frac{V(H)}{m}} \log \left[\frac{m}{V(H)} + \frac{1}{\delta}\right]\right)$$

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